Coherent Signal Processing for Loosely Coupled Bistatic Radar

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Sensitivity and performance of homodyne radar are notably improved if the phase noise terms of transmit and receive signals are correlated and canceled out by the receiver. However, in a bistatic setup with separate oscillators, this correlation is inherently not given. To achieve a good carrier phase and phase noise coherency, it is required to transmit a radio frequency (RF) reference, e.g., by using high-quality RF cables, optical fibers, or wireless point-to-point links. All these options are complex, costly, and prone to distortions and signal delay variations. Sharing a clock signal is also barely helpful since even small distortions of this reference are converted to notable uncorrelated phase noise in the radar signal synthesizers. In this article, a novel full-duplex system concept and signal processing scheme is presented that solves the above-mentioned problems completely. This concept allows for coherent measurements within a bistatic radar setup, even if the two radar stations are only loosely coupled employing a simple clock-rate adjustment. By theory and experimental results with bistatic frequency-modulated continuous-wave radar, it is shown that our concept allows for bistatic measurements with the same sensitivity and performance that is usually only known from homodyne radar systems. To the authors' best knowledge, this is the first time that phase noise coherent measurements in a bistatic multiple-input multiple-output radar setup are shown. This concept paves the way for advanced netted radar setups, e.g., in automotive radar and applications with large apertures or baselines or multiperspective coherent tomographic measurements.

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I. INTRODUCTION

Advanced driver-assistance systems or self-driving cars are equipped with multiple sensors and multiple sensor types. A typical sensor set includes, for example, four or eight radar modules for advanced safety functions [1]. Up to now, these multiple radar sensors mostly operate incoherently, i.e., separately with respect to the millimeter-wave radar signal. The data scanned by the distributed radar sensors are combined primarily on target list level or grid map level to form a 360° 2-D or 3-D perception of the car surroundings. The potential accuracy and resolution of the obtained images are determined by the radar signal bandwidth and the radar aperture size. The latter is defined by the antenna array size (essentially, as well, the size of the virtual aperture of a multiple-input multiple-output (MIMO) array) that can be twice the module size [2], or the size of a synthetic aperture created by the car's motion [3]. The applicable radar module size is usually limited by the printed circuit board cost and installation space constraints. Synthetic aperture radar (SAR) use cases require ultraprecise ego-motion estimation or complex autofocusing algorithms and are only effective for side-looking arrangements [3] and, thus, only slightly improve front- or rear-view images.

We know from personnel security screening systems [4] that large coherently operating radar networks can solve the above-mentioned accuracy and resolution issues. In a coherent radar network, the resolution is limited by the aperture size of the complete network and not by the module size. To apply this idea to automotive radar is challenging since it usually requires a common coherent carrier phase at all radar modules in the network. Practically, coherent measurements can be carried out with continuous phase synchronization by additional wireless radio frequency (RF) links. Such systems are known from satellite interferometric SAR and require expensive ultrastable oscillators [5]. As shown in [6] and [7], the use of optical fibers enables the operation with separated radar sensors. This concept employs additional optical and electro-optical components that are not part of leading-edge radar sensors.

A hardware setup that is frequently used for bistatic frequency-modulated continuous-wave (FMCW) radar sensors comparable to [8] is shown in Fig. 1. The only physical interconnections between the separate radar units are a clock rate adjustment link and a trigger cable that starts a chirp sequence. Without further processing steps, this setup does not allow phase coherent or phase noise coherent operation [9] and to take full advantage of MIMO operation. Based on this setup, coherent processing of combinations of receive (RX) and transmit (TX) channels, comprised of bistatic and monostatic responses, has been demonstrated in [10]. To this end, two radar units are connected by a trigger for single FMCW chirps and sampled data are transferred to a central processing unit. Frequency-division multiplexing (FDM) of the baseband signals of two radar stations can be realized by a delayed ramp start and/or a small offset in the carrier frequencies. In imaging radar, an evaluation

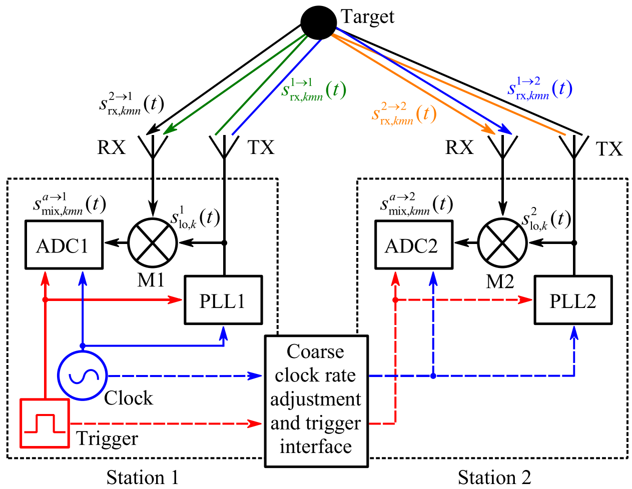


Fig. 1. Schematic overview of the setup with two FMCW radar stations with adjusted system clock sources and a trigger. Each radar station consists of PLLs as RF signal sources, frequency mixers, and analog-to-digital converters. There are at least one transmitting and one receiving antenna element per radar station. Signals, which are reflected by a passive target, are indicated with different colors to show distinct signal paths.

of such combinations is likely to create ghost targets that cause false positives. Especially in automotive scenarios with numerous targets, a linear method is required for the angular estimation to avoid this problem.

The application of a processing-based method related to this work to cooperative local positioning has been shown in [11]. Here, a combined evaluation of signal paths enhances the accuracy of the range estimation, mainly by the suppression of multipath propagation. Local positioning is typically based on a strong line-of-sight (LOS) transmission and does not require the detection and resolution of multiple passive targets. For such applications, velocity and angular estimation with MIMO radar units is not necessary and has not been demonstrated. Another application that uses reflections at a dielectric lens for the required phase calibration to enable coherent angular measurements is shown in [9]. This system setup has limited practical use as it cannot be employed for widely separated radar units. For two collocated arrays with numerous antenna elements, a practical implementation with a single lens is also demanding. Another application with single-channel radar units that form a single spatially distributed MIMO radar can be found in [12]. In multiple target scenarios, the position estimation via multilateration with single-channel units becomes challenging. In addition, the lack of angular discrimination leads to an uncertainty in the association of targets and, therefore, in the estimation of unknown variables, e.g., frequency and phase offsets. We solve this problem by coherent operation with two full MIMO radar units. We demonstrated in our previous work [13] that coherent measurements can be carried out together by separated incoherent radar stations using the coherent full-duplex double-sided two-way ranging method. After simultaneous transmission and reception of the RF signals (full-duplex), sampled data of both radar units (double-sided) are used to calculate offsets and archive phase and phase noise coherency. The presented algorithm for a single channel LOS transmission includes data-based synchronization steps that are required for wireless applications. To decrease complexity, timing and frequency offsets between two MIMO units are kept constant by using a setup comparable to [8] in this work.

Theoretical concepts and advantages of distributed MIMO radar systems have been investigated in [14]–[16]. Therein, the spatial diversity gain for radar targets with angle-dependent radar cross section (RCS) is explained for incoherent processing with widely separated antennas. Furthermore, centralized coherent processing yields increased resolution and enhanced angular accuracy. We provide a practical solution for both problems in this article. First, we describe the operation of widely separated state-of-the-art radar units by means of coherent velocity estimation (intrachirp coherence). Second, we examine a fully coherent configuration of two adjacent radar units with an increased aperture. We also derive a calibration technique in the near-field of the resulting virtual array and required processing steps based on a spatially matched filter.

The rest of this article is organized as follows. Section II provides an overview of the system setup and explains possible configurations and different operation modes. The signal model and coherent velocity estimation will be derived in Section III. Fully coherent operation with distributed radar units will be explained in Section IV in terms of a near-field reconstruction algorithm and an efficient calibration technique. Simulation and experimental results with two 77 GHz MIMO radar sensors will be provided in Sections V and VI to verify the developed theoretical concepts.

II. SYSTEM SETUP AND OPERATION MODES

This section provides an overview of hardware requirements and the data sampling principle and describes two possible operation modes.

A. System Setup and Measurement Principle

Without loss of generality, the following description explains coherent operation with two radar units. A measurement setup with three or more radar units utilizing an additional intermediate frequency (IF) bandwidth can be operated in a straightforward manner, as shown in [17]. The basic configuration, as depicted in Fig. 1, is composed of two FMCW radar units. Given that the RF signals are not distributed means expensive high-end cables (e.g., coaxial cables) are redundant. The RF signals are generated by a phase-locked loop (PLL1 and PLL2) with a stable reference clock (e.g., a quartz oscillator). The two radar units are only connected with a simple interface for coarse clock rate adaption and for triggering. In the used experimental setup, the interface is realized with a simple two-wire cable and an 80 MHz clock signal that is used to adjust the system clock of stations 1 and 2. These RF signals are denoted by $s^{a}_{\mathrm{lo},k}(t)$, where a indicates the radar station number, k indicates the FMCW chirp number, and the local oscillator (LO) defines the signal function. These signals are transmitted by an antenna, reflected at a target, and received as $s_{\mathrm{rx},kmn}^{a\to b}(t)$, where the transmission from station a to b is defined by the superscript $a \to b$ and the subscripts $m \in \{1,\ldots,M\}$ and $n \in \{1,\ldots,N\}$ define the RX and TX elements, respectively. A chirp sequence is comprised of K chirps; and there are M RX elements and N TX elements. We assume the same number of TX and RX elements at both stations. Down-conversion of $s_{\mathrm{lo},k}^{a}(t)$ is carried out by RF mixers (M1 and M2) and the resulting beat signals $s_{\mathrm{mix},kmn}^{a\to 1}(t)$ and $s_{\mathrm{mix},kmn}^{2\to 1}(t)$ are respectively sampled by analog-to-digital-converters ADC1 and ADC2. For the sake of simplicity, additional components, such as low-pass filters and amplifiers, are omitted in Fig. 1 and a single-input single-output system is depicted. For operation with a larger real and/or virtual aperture, MIMO radar systems are employed.

Each chirp sequence is triggered by a dedicated physical signal to start the measurement sequence with a fixed time delay $\Delta\tau$ between station 1 and station 2. After triggering, a sequence of K FMCW chirps is transmitted and received simultaneously at both stations in full-duplex mode. Separation of homodyne/monostatic and heterodyne/bistatic signals is enabled by a defined offset $\Delta\tau$ in time and/or Δf in frequency. As shown in [10], the use of a trigger signal ensures that the random initial phases of the down-converted heterodyne mixing products of single FMCW chirps are identical at both stations. As the timing is defined by the system clock, clock rate adjustment extends this property to all chirps in a sequence and the target velocity is estimated with spatially separated radar units after appropriate correction steps.

Additionally, the carrier frequency is derived from the system clock. Incoherent oscillators have a certain relative frequency drift, which influences the RF carrier and causes a phase shift depending on the time-of-flight (ToF). This is caused by imprecise fabrication processes, thermal conditions, aging, etc. [18]. Adjusted clocks avoid these problems and the reconstructed spectrograms from both radar stations can be evaluated in an interferometric way to increase the array aperture. In this system setup, the RF carrier at 77 GHz is generated by separate PLLs from a stable 80 MHz reference, which is distributed via loose coupling. As described in [19], frequency multiplication by a factor of n increases the spectral phase noise density by n^2 (here: approx. 60 dB) independently in both stations. With this configuration, the phase noise of stations 1 and 2 can be assumed to be highly uncorrelated and fast chirp rates are needed to guarantee sufficiently low distortions [20].

B. Operation Modes

In principle, the exploitation of an RCS that depends on the aspect angle (angle-dependent RCS) and vectoral velocity estimation both benefits from a multiperspective view with a large distance (for instance, 1 m) between radar stations [15]. In this case, the resulting interferogram shows a high sidelobe level (SLL) because the resulting antenna array is very sparse and target detection becomes infeasible. Antenna arrays with this property are known from radio astronomy and take advantage of the earth's rotation to generate a densely occupied array [21]. Alternatively, the spatial separation must be small enough (e.g., 10 cm) to enable an SLL which is sufficiently low. As both applications require a different physical arrangement and corresponding processing steps, these two operation modes will be distinguished.

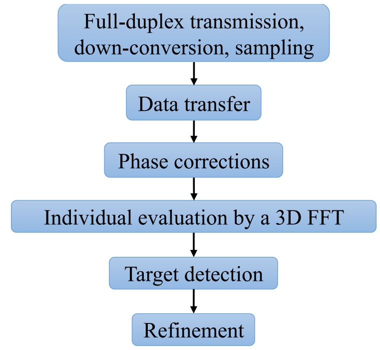


Fig. 2. FMCW chirp sequence is simultaneously transmitted and received at both stations for coherent chirp-sequence processing with distributed units. After the sampled signals are transferred, phase distortions are compensated to enable a 3-D FFT and range and velocity can be estimated accurately by a subsequent refinement.

1) *Coherent Chirp-Sequence Processing:* An overview of necessary steps in the operation mode with large spatial separation between two radar stations is given in sequential order in Fig. 2. The transmission scheme according to Section II-A has been executed and the mixing products have been digitally sampled, ADC data from both radar units are transferred either to a single radar station or to a master processing unit. Thereafter, a 1-D fast Fourier transform (FFT) along range dimension is computed and the phase of the strongest target or multiple strong targets is determined for every RX channel and FMCW chirp in both bistatic responses. As the phase perturbations are the same in both stations, these variations can be determined and corrected along a chirp sequence.

Hereafter, two additional three-dimensional range-Doppler-azimuth spectra for the bistatic signals become available and benefit from coherent signal-to-noise ratio (SNR) gain. The target position can then be estimated with increased SNR and the bistatic Doppler shift. Finally, targets are detected and assigned to compensate for fast time phase noise and Doppler phase errors. The resulting accuracy becomes comparable to a homodyne FMCW radar.

2) *Fully Coherent Operation:* For increased angular resolution and accuracy with sufficiently low SLL, a setup with a smaller distance between the radar stations is the preferred choice [15]. The flowchart in Fig. 3 shows the necessary processing steps for a colocated setup, which are the same as in Fig. 2 through to the transfer of ADC data. Afterward, the monostatic responses are reconstructed with a spatially matched filter to yield an interferogram with high angular resolution. Geometrical effects on the phase of the bistatic responses must be compensated for every FMCW chirp, given that the initial phases of both stations need to

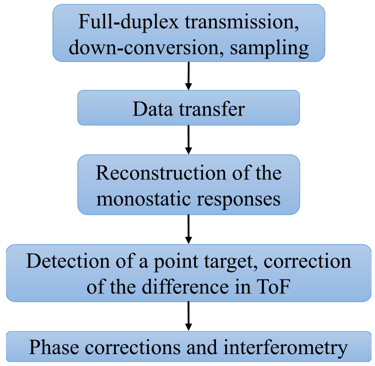


Fig. 3. In the fully coherent operation mode, the monostatic responses from both stations are reconstructed to create an interferogram. This interferogram is used to detect a point target. This difference of the ToFs is corrected in order to estimate phase distortions. After these distortions have been compensated, interferometry is applied, utilizing all virtual channels to create a larger aperture.

remain constant to reconstruct all virtual channels. A strong isolated point target is, therefore, required as the basis for good phase perturbation estimation and correction. Finally, an interferogram is computed with all virtual channels. The result is comparable to that of a single MIMO radar with a large aperture. Additionally, the radial velocity of objects can be determined for every pixel by a 1-D FFT in a slow time dimension.

III. SIGNAL MODEL AND COHERENT VELOCITY ESTIMATION

Initially, this section describes the signal model of the bistatic response, introduces the corresponding geometry, and explains all assumptions. Thereafter, coherent velocity estimation with a large spatial separation will be explained.

A. Signal Model

Based on [22], a description of the bistatic radar response with single FMCW chirps can be found in [10]. This concept will be extended to a chirp sequence, which is comprised of K chirps. For the sake of simplicity, all calculations are carried out in the continuous-time domain. The LO signal of station 1 with normalized magnitude can be given as

$$s^1_{lo,k}(t) = \exp\left\{ j\left(2\pi\left(f_c t + \mu t^2/2\right) + \Psi^1_{pn,k}(t) + \varphi^1_k\right)\right\} \tag{1}$$

where f_c, $\Psi^1_{pn,k}(t)$, and φ^1_k denote RF carrier frequency, phase noise at station 1, and an unknown initial phase, respectively. The variable $\mu = B/T_{sw}$ is defined as the chirp rate for a linear chirp with bandwidth B and a sweep time of T_{sw}.

Station 2 operates with either a positive or negative offset in frequency and/or time Δf and $\Delta\tau$, respectively. These offsets ensure that monostatic and bistatic responses occupy distinguishable parts of the IF spectrum. In practice, the use of one of these parameters is enough. For a system made up of two stations, the IF spectrum can be divided into three equal parts [17]. Therefore, Δf and $\Delta\tau$ are chosen in the range of a few megahertz and nanoseconds, respectively. The second LO signal is given as

$$\begin{aligned} s^2_{lo,k}(t) = \exp\Big\{ j\Big(2\pi\left(\left(f_c - \Delta f\right)(t-\Delta\tau) + \mu(t-\Delta\tau)^2/2\right) \\ + \Psi^2_{pn,k}(t-\Delta\tau) + \varphi^2_k\Big)\Big\}. \end{aligned} \tag{2}$$

As the LO signal of an FMCW radar serves as a reference for the down-conversion and as a transmit signal, (1) and (2) are radiated by the corresponding TX antenna. After reflection at a target, a delayed and attenuated replica of the transmit signal can be defined as

$$s^{a\to b}_{rx,kmn}(t) = \alpha^{a\to b} s^a_{lo}(t - \tau^{a\to b}_{kmn}) \exp\{j2\pi f_D t\} \tag{3}$$

where $\alpha^{a\to b} \in \mathbb{C}$ indicates the complex attenuation coefficient, which depends on the signal path from station a to station b. The variable $\tau^{a\to b}_{kmn}$ denotes the ToF, which is based on the geometry, i.e., the position of a TX element n at station a, an RX element m at station b, and on the chirp number to account for the velocity of a target. The Doppler frequency f_D related to $f_c + B/2$ is assumed to be the same at both stations, as the frequency offset is negligible compared to the RF carrier frequency. During an FMCW chirp, the change in the target position can be neglected by selecting adequate chirp parameters or compensated, as described in [23].

Based on these considerations, the mixing products of the bistatic response are now modeled as complex functions with a positive frequency. The mixing product of the bistatic signals in (1) and (3) at station 1 can be calculated as

$$\begin{aligned} s^{2\to1}_{mix1,kmn}(t) &= s^1_{lo,k}(t)\left(s^{2\to1}_{rx,kmn}(t)\right)^* \\ &= \alpha^{2\to1}\exp\left\{j2\pi t\left(\Delta f + \mu\Delta\tau + \mu\tau^{2\to1}_{kmn} - f_D\right)\right\} \\ &\quad\cdot \exp\Big\{j\Big(2\pi\left(f_c\Delta\tau - \Delta f\Delta\tau + f_c\tau^{2\to1}_{kmn} - \Delta f\tau^{2\to1}_{kmn}\right) \\ &\quad + \Psi^1_{pn,k}(t) - \Psi^2_{pn,k}\left(t - \Delta\tau - \tau^{2\to1}_{kmn}\right) + \varphi^1_k - \varphi^2_k\Big)\Big\} \end{aligned} \tag{4}$$

where quadratic terms of $\tau^{2\to1}_{kmn}$ and $\Delta\tau$ can be neglected for typical distances and sufficiently low chirp rates. The first line after the second equality sign contains all variables that mainly affect the beat frequency. The second and third lines show all radar parameters that define the signal phase. Analogously, the mixing product of (2) and (3) at the second radar station sampled at the local time $t + \Delta\tau$ can be calculated as

$$\begin{aligned} s^{1\to2}_{mix2,kmn}(t) &= s^{1\to2}_{rx,kmn}(t+\Delta\tau)\left(s^2_{lo,k}(t+\Delta\tau)\right)^* \\ &= \alpha^{1\to2}\exp\left\{j2\pi t\left(\Delta f + \mu\Delta\tau - \mu\tau^{1\to2}_{kmn} + f_D\right)\right\} \\ &\quad\cdot \exp\Big\{j\Big(2\pi\left(f_c\Delta\tau - f_c\tau^{1\to2}_{kmn}\right) \\ &\quad + \Psi^1_{pn,k}\left(t + \Delta\tau - \tau^{1\to2}_{kmn}\right) - \Psi^2_{pn,k}(t) + \varphi^1_k - \varphi^2_k\Big)\Big\}. \end{aligned} \tag{5}$$

Comparing (4) and (5) reveals that both beat signals are centered around the frequency $f_{center} = \Delta f + \mu\Delta\tau$, as shown separately for offsets in frequency [17] and time [10]. The following section explains the relationship between (4) and (5), the determination of unknown parameters, and

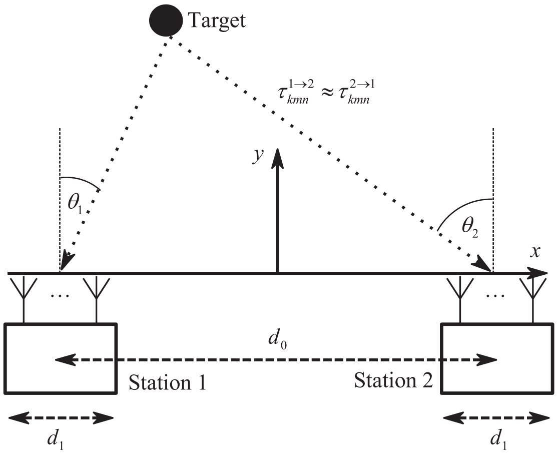


Fig. 4. Setup based on [24], which exploits an angle-dependent RCS and enables the velocity estimation with bistatic signals, requires a sufficiently large separation between both radar units. As the geometric dimensions d_1 of a single radar are small compared to d_0, additional target parameters can be determined which are almost identical in all channels of a specific unit.

the reconstruction of 3-D range-Doppler-azimuth spectrograms.

B. Coherent Velocity Estimation

The basic bistatic radar setup with a single TX and RX antenna is described in [24]. An extended configuration with two MIMO radar stations is given in Fig. 4. The geometrical dimension of one radar is d_1 and the spatial separation between two radar stations is d_0. As the resulting aperture of the virtual array depends on the aspect angle, it cannot be defined readily but its maximum size can be given as $2d_1$. To benefit from an angle-dependent RCS, $d_0 \gg 2d_1$ must be fulfilled. Therefore, the azimuth angles θ_1 and θ_2 are different but the bistatic ToF is almost the same in both stations. The velocity offset between different TX and RX channels of a MIMO radar can be neglected if the aperture size is small. The global 2-D Cartesian coordinate system is defined by x-y coordinates. The geometrical relationship between two bistatic range-Doppler-azimuth spectrograms in local polar coordinates—which will be determined in this section—and the global coordinate system is explained in [25].

1) *Assumptions:* To avoid additional correction steps in this section, the mathematical model is based on a single TX element at each station, i.e., $N = 1$. Therefore, the index n is omitted here and $\tau_{kmn}^{a\to b}$ becomes $\tau_{km}^{a\to b}$ for a SIMO system. The processing steps with multiple TX channels will be explained in Section IV. The signal path attenuation from station 1 to station 2 and vice versa is assumed to be comparable, i.e., $\alpha^{1\to 2} \approx \alpha^{2\to 1}$. The ToF

$$\tau_{km}^{a\to b} = \tau_0 + \tau_{\mathrm{v},k} + \tau_{\mathrm{a},m}^{a\to b} \tag{6}$$

of the bistatic response is comprised of the initial ToF, the change of ToF due to velocity, and the angle-dependent component, which are τ_0, $\tau_{\mathrm{v},k}$, and $\tau_{\mathrm{a},m}^{a\to b}$, respectively. As is typical for FMCW radar with fast chirp rates, the influence of these terms on each other is negligible [26].

The intrachirp phase noise must be sufficiently small to detect targets after the FFT along the fast-time dimension. This can be achieved with high-quality RF sources in combination with fast chirp rates [20]. Furthermore, the realizations of the random walk of the phase deviation in stations 1 and 2 are highly correlated for a relatively short ToF [27]. Therefore, the perturbation caused by phase noise is very similar in both stations and can be included in the random initial phase as $\tilde{\varphi}_k^1 - \tilde{\varphi}_k^2$. We shall describe how the remaining phase errors are offset along fast and slow time dimensions in 3).

Different from local positioning use cases, the offsets Δf and $\Delta\tau$ are time-invariant due to the distribution of clock and trigger signals. If the offsets cannot be preset accurately (here, the length of the trigger cable causes an additional time delay), these values can be determined independently by calibration, e.g., by employing up- and down-chirps [28].

2) *Estimation and Correction of Unknown Parameters:* Based on the assumptions above and the Fourier transform along the fast time dimension, the station 1 mixing product in (4) can be simplified to

$$\begin{aligned} S_{\mathrm{mix},km}^{2\to 1}(f) &= \mathcal{F}\left\{w(t)s_{\mathrm{mix},km1}^{2\to 1}(t)\right\} \\ &= \alpha^{2\to 1}W\left(f - \left(f_{\mathrm{center}} + \mu\tau_{km}^{2\to 1} - f_{\mathrm{D}}\right)\right) \\ &\quad \cdot \exp\left\{j\left(\vartheta_k + 2\pi\left(f_{\mathrm{c}}\tau_{km}^{2\to 1} - \Delta f\tau_{km}^{2\to 1}\right)\right)\right\} \end{aligned} \tag{7}$$

where the random phase for each chirp is modeled as $\vartheta_k = 2\pi(f_{\mathrm{c}} - \Delta f)\Delta\tau + \tilde{\varphi}_k^1 - \tilde{\varphi}_k^2$. The functions $w(t)$ and $W(f)$ denote the window functions in the time and frequency domain, respectively. The $\mathcal{F}\{\cdot\}$ operator denotes the Fourier transform. The first line after the second equality sign represents the beat frequency and the next line the signal phase. Accordingly, the spectrum of the second mixing product in (5) after compensating for the time delay can be defined as

$$\begin{aligned} S_{\mathrm{mix},km}^{1\to 2}(f) &= \mathcal{F}\left\{w(t)s_{\mathrm{mix},km1}^{1\to 2}(t - \Delta\tau)\right\} \\ &\approx \alpha^{1\to 2}W\left(f - \left(f_{\mathrm{center}} - \mu\tau_{km}^{1\to 2} + f_{\mathrm{D}}\right)\right) \\ &\quad \cdot \exp\left\{j\left(\vartheta_k - 2\pi f_{\mathrm{c}}\tau_{km}^{1\to 2}\right)\right\}. \end{aligned} \tag{8}$$

The value f_{center} can be determined by comparing the last two equations. Hereafter, the influence of the frequency offset on the signal phase in (7) can be compensated for by multiplying by $\exp(j2\pi\Delta f\tau_{km}^{2\to 1})$. Alternatively, this offset could be used for operation as a frequency diverse array [29].

After these steps with the relationship (6), the similarities between the beat signals

$$\begin{aligned} \widehat{S}_{\mathrm{mix},km}^{2\to 1}(f) &\approx \alpha^{2\to 1}W\left(f - \mu\tau_0 + f_{\mathrm{D}}\right) \\ &\quad \cdot \exp\left\{j\left(\vartheta_k + 2\pi f_{\mathrm{c}}\left(\tau_{\mathrm{v},k} + \tau_{\mathrm{a},m}^{2\to 1}\right)\right)\right\} \end{aligned} \tag{9}$$

$$\begin{aligned} \widehat{S}_{\mathrm{mix},km}^{1\to 2}(f) &\approx \alpha^{1\to 2}W\left(f - \mu\tau_0 + f_{\mathrm{D}}\right) \\ &\quad \cdot \exp\left\{j\left(\vartheta_k - 2\pi f_{\mathrm{c}}\left(\tau_{\mathrm{v},k} + \tau_{\mathrm{a},m}^{1\to 2}\right)\right)\right\} \end{aligned} \tag{10}$$

at the first and second station, respectively, become clear. The bistatic signal at station 2 in (10) was mirrored at

f_{center}. For each FMCW chirp, these signals are influenced by the unknown phase value ϑ_k, which is independent of the RX channel and interferes with the phase that corresponds with the target velocities. Therefore, ϑ_k is determined by comparing the phase of the strongest peak of both beat spectra

$$\begin{aligned}\phi_{km}^{1} &= \arg\left\{\widehat{S}_{\text{mix},km}^{2\to 1}\left(\arg\max_{f}\left\{\left|\widehat{S}_{\text{mix},km}^{2\to 1}(f)\right|\right\}\right)\right\} \\ &= \vartheta_k + 2\pi f_{\text{c}}\left(\tau_{\text{v},k} + \tau_{\text{a},m}^{2\to 1}\right) \end{aligned} \quad (11)$$

for each RX channel at station 1 and the phase values

$$\begin{aligned}\phi_{km}^{2} &= \arg\left\{\widehat{S}_{\text{mix},km}^{1\to 2}\left(\arg\max_{f}\left\{\left|\widehat{S}_{\text{mix},km}^{1\to 2}(f)\right|\right\}\right)\right\} \\ &= \vartheta_k - 2\pi f_{\text{c}}\left(\tau_{\text{v},k} + \tau_{\text{a},m}^{1\to 2}\right) \end{aligned} \quad (12)$$

at station 2, where the absolute value, the operator that returns the argument of the maximum, and the argument of a function are denoted by $|\cdot|$, $\arg\max\{\cdot\}$, and $\arg\{\cdot\}$, respectively. The phase correction value can be found by individually unwrapping and averaging the phase values in (11) and (12) for each FMCW chirp. Mathematically, this operation can be defined as

$$\phi_{k}^{1/2} = \frac{1}{M}\sum_{m=1}^{M} U\left\{\phi_{km}^{1/2}\right\} = \vartheta_k \pm 2\pi f_{\text{c}}\tau_{\text{v},k} + \phi_{0}^{1/2} \quad (13)$$

where the $U\{\cdot\}$ operator denotes phase unwrapping along M RX channels. The $\phi_0^{1/2}$ variable accounts for an unknown phase value that depends on the geometry. The averaging over m in (13) increases the SNR, which can be improved further by including more targets. Thereafter, the mean value

$$\varphi_{0,k} = \frac{\phi_k^1 + \phi_k^2}{2} = \vartheta_k + \hat{\phi}_{0,k} \quad (14)$$

of the phase estimates at stations 1 and 2 in (13) is comprised of the phase perturbation of each FMCW chirp ϑ_k and the variable $\hat{\phi}_{0,k}$, which accounts for the reduced unambiguous range due to the division by two and the geometry. By the aid of (14), the beat spectra in (9) and (10) are corrected by multiplying by $\exp(-\mathrm{j}\varphi_{0,k})$ and subsequent unwrapping. Hence, the phase shift due to velocity in (9) and (10) must not exceed $\pm\pi/2$.

The resulting beat spectra can be given as

$$\begin{aligned}\widehat{\widehat{S}}_{\text{mix},km}^{2\to 1}(f) &= \alpha^{2\to 1} W\left(f - \mu\tau_0 + f_{\text{D}}\right) \\ &\cdot \exp\left\{j\left(2\pi f_{\text{c}}\left(\tau_{\text{v},k} + \tau_{\text{a},m}^{2\to 1}\right) - \hat{\phi}_{0,1}\right)\right\} \end{aligned} \quad (15)$$

at station 1 and

$$\begin{aligned}\widehat{\widehat{S}}_{\text{mix},km}^{1\to 2}(f) &= \alpha^{1\to 2} W\left(f - \mu\tau_0 + f_{\text{D}}\right) \\ &\cdot \exp\left\{j\left(2\pi f_{\text{c}}\left(\tau_{\text{v},k} + \tau_{\text{a},m}^{1\to 2}\right) + \hat{\phi}_{0,1}\right)\right\} \end{aligned} \quad (16)$$

at station 2. Complex conjugation in (16) results in the same sign for moving targets as in (15). The phases of these signals are a function of the target's angle and velocity. Except for an unknown initial phase, these signals are comparable to a monostatic radar response. Both stations have the same Doppler phases but different aspect angles due to the bistatic radar system geometry. Finally, an FFT along slow time dimension is computed and two range-Doppler-azimuth diagrams are generated by subsequent RX-beamforming. The relationship between the ToF in (6) and target parameters is explained in [26]. The influence of the Doppler frequency on the ToF can be corrected after the Doppler phase shift has been evaluated.

The additional random phase of each chirp sequence renders the calibration of RX phases by averaging impossible. Therefore, the RX antenna array phases can be calibrated with the technique described in Section IV-C. Another option is to employ the method described in [30], which uses the phase differences instead of the absolute phases. Creating an interferogram causes very high sidelobes and is impractical, given the large gap between both stations.

3) *Refinement of Ranging and Velocity Accuracy:* Up to this point, we have neglected the influence of phase noise along the fast time dimension on the estimation performance of the unknown parameters in 2), i.e., ranging and bistatic velocity. Intrachirp PN reduces the bistatic ranging accuracy for targets with a high RCS (in this case, PN is the dominant source of error). This effect can be compensated by a linear combination of both ranging results, which allows the compensation of phase noise along the fast time dimension, and FMCW signal nonlinearities [27]. Having said that, the phase estimation also suffers from the effect of fast time PN on the detected unknown initial phases in (11) and (12). Hence, the computed target velocities are less accurate than with a homodyne FMCW radar. These small phase errors can be approximated as a linear function via Taylor expansion [31].

In contrast to localization applications with a prominent LOS signal, targets in (15) and (16) must be detected and assigned. Practically, the results of a 3-D FFT of (15) and (16) can be transformed into a global Cartesian coordinate system (see [25]) and targets can be identified at the same location with almost the same velocity.

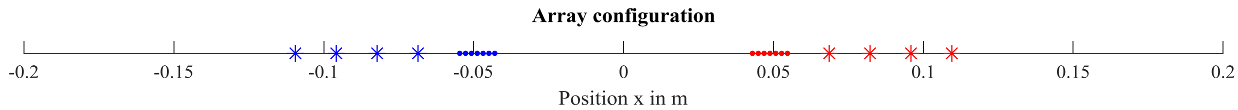
Hence, the imprecise ranging and velocity results in both directions of transmission can be modeled as

$$\begin{pmatrix}\hat{\tau}_0^{a\to b} \\ \hat{\tau}_{\text{v}}^{a\to b}\end{pmatrix} = \begin{pmatrix}\tau_0 \pm \tau_{0,\varepsilon} \\ \tau_{\text{v}} \pm \tau_{\text{v},\varepsilon}\end{pmatrix} \quad (17)$$

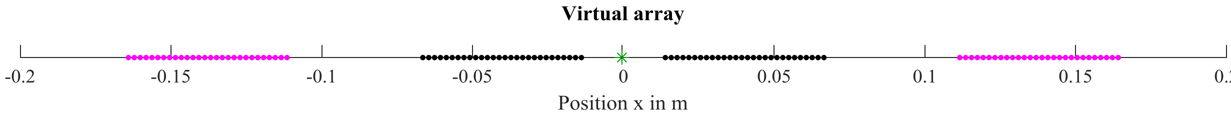
where the variables $\tau_{0,\varepsilon}$ and $\tau_{\text{v},\varepsilon}$ denote the ToF estimation error and the change in ToF due to the velocity of a target, respectively. Finally, accurate bistatic estimation results

$$\begin{aligned}\hat{\tau}_0 &= \left(\hat{\tau}_0^{2\to 1} + \hat{\tau}_0^{1\to 2}\right)/2 \quad \text{and} \\ \hat{\tau}_{\text{v}} &= \left(\hat{\tau}_{\text{v}}^{2\to 1} + \hat{\tau}_{\text{v}}^{1\to 2}\right)/2 \end{aligned} \quad (18)$$

can be computed as the mean value of stations 1 and 2. The effect of phase noise on the RX channels and the angular estimation can be neglected as the difference in ToF is very small.



(a)



(b)

Fig. 5. (a) Configuration with two radar stations close to each other. Here, $N = 4$ TX elements are marked as crosses and $M = 7$ RX elements as dots. Station 1 is colored in blue and station 2 in red and the distance between the inner RX elements is $d_2 = 8.6$ cm. (b) Resulting virtual array, where elements in magenta depict the resulting element positions from the combined estimation with the monostatic responses. Black elements show the combination of the bistatic responses and the virtual TX element is marked as the green cross.

IV. FULLY COHERENT OPERATION AND CALIBRATION

Different from Section III, this section describes the joint reconstruction with all monostatic and bistatic responses to create a single radar image. Hence, the resulting virtual array has a larger aperture and more virtual elements. More specifically, the complete signal model will be given in Section IV-A. In Section IV-B, reconstruction with a spatially matched filter will be explained and Section IV-C introduces a practical calibration technique for the bistatic signals.

A. Signal Model

To avoid a high SLL caused by a large gap between two radar stations, a setup with small spatial separation is favorable. An example with a distance of $d_2 = 8.6$ cm between the inner RX elements is depicted in Fig. 5. Both MIMO radar stations in Fig. 5(a) are identical and use $N = 4$ TX elements and $M = 7$ RX elements, which form a uniform linear array (ULA) with $\lambda/2$ spacing in MIMO mode. The positions of the TX and RX elements at station a are denoted by $\mathbf{x}^a_{\mathrm{TX},n}$ and $\mathbf{x}^a_{\mathrm{RX},m}$, respectively. The virtual array in Fig. 5(b) shows the extended aperture enabled by coherent signal processing with both radar stations. It is comprised of four ULAs, which are separated by three gaps. As the new aperture size is $d_3 = 33$ cm, the far-field condition for a 77 GHz system holds for targets at ranges greater than $2d_3^2/\lambda \approx 55.9$ m, where λ denotes the wavelength of the RF carrier [32]. Especially for automotive applications, target detection close to the vehicle is necessary. In this region, the plane wave assumption is invalid and the target angle cannot be estimated by an FFT or digital beamforming of signals with a large RF bandwidth. With the aid of a spatially matched filter, the reconstruction can be done without a plane wave assumption [33]. A fully coherent operation mode with all virtual channels is not viable with the phase correction in Section III, as the initial phase $\hat{\phi}_{0,1}$ in (15) and (16) is unknown. Therefore, the coordinates of a point target must be determined and the geometrical impact must be compensated prior to the bistatic phase difference estimation. Different from the previous section, the reflection coefficients along individual signal paths are almost the same for a small inter-station distance and are modeled as $\alpha \in \mathbb{C}$ for all monostatic and bistatic responses.

Well-known multiplexing schemes for MIMO radar systems that ensure orthogonality are time-division multiplexing (TDM) and FDM [34]. The former approach requires the compensation of motion-induced errors, as explained in [35] and [36]. Alternatively, waveform multiplexing (e.g., FMCW chirps with different slopes) can be employed [37]. The TDM mode, whereby an unknown phase of the bistatic signals is corrected for every TX element, is explained next. Targets must be static in the TDM mode. Moreover, the index k will be dropped as the detection is carried out individually for every FMCW chirp. We shall then discuss the compensation of the interchirp Doppler frequency shift f_{D}.

The monostatic responses can be modeled as

$$S^{1\to 1}_{\mathrm{mix},mn}(f) = \alpha W\left(f - \mu\tau^{1\to 1}_{mn}\right)\exp\left(j2\pi f_{\mathrm{c}}\tau^{1\to 1}_{mn}\right) \quad (19)$$

where the superscript in $\tau^{1\to 1}_{mn}$ denotes that signals are transmitted by the first station, reflected at an object, and received by the first station as well [22]. In the case of a homodyne mixing process and sufficiently short ranges, the influence of phase noise can be omitted due to its high correlation in TX and RX signals. The signal phase of the second monostatic response

$$S^{2\to 2}_{\mathrm{mix},mn}(f) = \alpha W\left(f - \mu\tau^{2\to 2}_{mn}\right)\exp\left(j2\pi\left(f_{\mathrm{c}} - \Delta f\right)\tau^{2\to 2}_{mn}\right) \quad (20)$$

depends on the frequency offset Δf. An adjusted clock signal (see Fig. 1) guarantees that both stations' RF carriers are identical except for this offset.

After compensating the frequency offset $f_{\mathrm{center}} = \Delta f + \mu\Delta\tau$ of the bistatic responses, as explained in Section III-B2), the first station's beat signal can be expressed as

$$S^{2\to 1}_{\mathrm{mix},mn}(f) = \alpha W\left(f - \mu\tau^{2\to 1}_{mn}\right) \cdot \exp\left(j\left(\vartheta_n + 2\pi\left(f_{\mathrm{c}} - \Delta f\right)\tau^{2\to 1}_{mn}\right)\right) \quad (21)$$

where the index n in ϑ_n accounts for the dependence of a random phase on the TX element n in TDM mode. The bistatic response at the second station after the correction steps in Section III-B2) and complex conjugation can be modeled as

$$S^{1\to 2}_{\mathrm{mix},mn}(f) = \alpha W\left(f - \mu\tau^{1\to 2}_{mn}\right) \cdot \exp\left(j\left(-\vartheta_n + 2\pi f_{\mathrm{c}}\tau^{1\to 2}_{mn}\right)\right). \tag{22}$$

An equalization is necessary before ϑ_n is estimated, given that all phase values in (21) and (22) are different. This will be done next by focusing on a point target. This operation is skipped and the phase correction is simplified in the case of a common TX/RX channel.

B. Spatially Matched Filter Reconstruction

Generally, the spatially matched filter reconstruction of the FMCW beat spectra $S^{a\to b}_{\mathrm{mix},mn}(f)$ after range compression of the time domain beat signals can be done by

$$I^{a\to b}(x,y) = \sum_{m=1}^{M}\sum_{n=1}^{N} S^{a\to b}_{\mathrm{mix},mn}(\mu\widehat{\tau}^{a\to b}_{mn}(x,y)) \cdot \exp\left(-\mathrm{j}2\pi f^{a}_{\mathrm{c}}\widehat{\tau}^{a\to b}_{mn}(x,y)\right) \tag{23}$$

where $\widehat{\tau}^{a\to b}_{mn}(x,y)$ denotes the hypothetical ToF corresponding to a point in 2-D given by the coordinates (x,y), and f^{a}_{c} expresses the RF carrier frequencies, i.e., $f^{1}_{\mathrm{c}} = f_{\mathrm{c}} + B/2$ and $f^{2}_{\mathrm{c}} = f_{\mathrm{c}} + B/2 - \Delta f$, for the transmissions of stations 1 and 2 [33]. The reconstruction via (23) is a coherent summation of all TX and RX combinations. If the hypothetical ToF corresponds to the ToF to a target, a peak can be detected at this location. The hypothetical ToF is defined geometrically by

$$\widehat{\tau}^{a\to b}_{mn}(x,y) = \left(\left\|\mathbf{x}^{a}_{\mathrm{TX},n} - \widehat{\mathbf{x}}_{\mathrm{T}}\right\| + \left\|\mathbf{x}^{b}_{\mathrm{RX},m} - \widehat{\mathbf{x}}_{\mathrm{T}}\right\|\right)\Big/ c_0 \tag{24}$$

where $\|\cdot\|$, $\widehat{\mathbf{x}}_{\mathrm{T}} = (x,y)^{\mathrm{T}}$, and c_0 denote the Euclidian norm, the hypothetical target location, and the phase velocity, respectively. In the case of a ULA, the computational complexity of (23) can be reduced by an FFT, as known from airborne SAR [38].

The interferogram of both monostatic responses can be calculated readily. Thereafter, the estimated position of a point target needs to be identified by

$$\mathbf{x}_{\mathrm{T,est}} = \arg\max_{x,y}\left\{\left|I^{1\to 1}(x,y) + I^{2\to 2}(x,y)\right|\right\}. \tag{25}$$

The interferometric approach in (25) yields a highly accurate angular estimation. We use this target's response to compensate for the unknown phase values in (21) and (22). To this end, the phase shift of the bistatic responses with $a \neq b$, caused by the positional difference of TX and RX elements, needs to be corrected. In TDM mode, the reconstruction at the estimated target position is done separately for every TX element by

$$\arg\left\{I^{a\to b}(\mathbf{x}_{\mathrm{T,est}})\right\} = \arg\left\{\sum_{m=1}^{M} S^{a\to b}_{\mathrm{mix},mn}(\mu\widehat{\tau}^{a\to b}_{mn}(\mathbf{x}_{\mathrm{T,est}})) \cdot \exp\left(-\mathrm{j}2\pi f^{a}_{\mathrm{c}}\widehat{\tau}^{a\to b}_{mn}(\mathbf{x}_{\mathrm{T,est}})\right)\right\} \approx \varsigma_0 \pm \vartheta_n, \tag{26}$$

where ς_0 denotes an arbitrary phase value that is independent of the TX Element. Equation (26) returns the argument of the complex reconstruction result of the bistatic response at the location in (25). The variable ς_0 considers that a target's range cannot be estimated at the carrier wavelength scale.

The phase differences

$$\hat{\vartheta}_n = \left(\arg\left\{I^{2\to 1}(\mathbf{x}_{\mathrm{T,est}})\right\} - \arg\left\{I^{1\to 2}(\mathbf{x}_{\mathrm{T,est}})\right\}\right)/2 \tag{27}$$

can be used to correct the bistatic target responses in (21) and (22). Again, there is an ambiguity caused by the division by 2 that needs to be corrected. As the phase of all individual target responses should be the same at the target location, constructive interference is expected from the superposition of an arbitrary bistatic and an arbitrary monostatic response at this point. Hence, the ambiguity in (27) can be resolved by evaluating the magnitude of

$$A_{\pm} = \left|I^{a\to a}(\mathbf{x}_{\mathrm{T,est}}) \pm I^{a\to b}(\mathbf{x}_{\mathrm{T,est}})\right|. \tag{28}$$

If the value A_- is larger than A_+, π needs to be added to the phase estimate in (27). In total, four combinations of bistatic and monostatic responses can be used for the correction. Thereafter, the corrected beat spectra of the first station

$$\begin{aligned}\tilde{S}^{2\to 1}_{\mathrm{mix},mn}(f) &= S^{2\to 1}_{\mathrm{mix},mn}(f)\cdot\exp\left(-j\hat{\vartheta}'_n\right)\\ &= \alpha W\left(f - \mu\tau^{2\to 1}_{mn}\right)\cdot\exp\left(j2\pi\left(f_{\mathrm{c}} - \Delta f\right)\tau^{2\to 1}_{mn}\right)\end{aligned} \tag{29}$$

and the second station

$$\begin{aligned}\tilde{S}^{1\to 2}_{\mathrm{mix},mn}(f) &= S^{1\to 2}_{\mathrm{mix},mn}(f)\cdot\exp\left(j\hat{\vartheta}'_n\right)\\ &= \alpha W\left(f - \mu\tau^{1\to 2}_{mn}\right)]\cdot\exp\left(j2\pi f_{\mathrm{c}}\tau^{1\to 2}_{mn}\right)\end{aligned} \tag{30}$$

are comparable with the monostatic response, where $\hat{\vartheta}'_n$ denotes the unambiguous phase estimate via (27) and (28). The phases in (29) and (30) only depend on the geometry, and the interferogram with all virtual channels can be reconstructed for every pixel by

$$I(x,y) = \left|I^{1\to 1}(x,y) + I^{2\to 2}(x,y) + \tilde{I}^{2\to 1}(x,y) + \tilde{I}^{1\to 2}(x,y)\right| \tag{31}$$

where $\tilde{I}^{2\to 1}(x,y)$ and $\tilde{I}^{1\to 2}(x,y)$ use the bistatic results in (29) and (30), respectively. The additional virtual channels enable the SLL reduction. Additionally, range and angle can be estimated more accurately due to the increased SNR.

If the MIMO radar operates with FDM, the phase perturbation is the same for all TX elements and only a single correction per FMCW chirp is required. Hence, (31) can be applied for every chirp to create the sequence $I_k(x,y)$ without calculating the absolute value. Moving targets can thus be detected with an FFT for every pixel after the spatially matched filter is applied [39]. Different from Section III, the combined evaluation delivers a single estimate for the radial velocity. Analogously to Section III-B, the influence of the Doppler effect on the beat frequency can be compensated by subtracting the associated estimated phase shift. The spatially matched filter based on (23) also allows a real and a synthetic aperture to be combined.

As explained in Section III.B.3), the bistatic ranging accuracy is impaired by phase noise along the fast time dimension. By the aid of the spatially matched filter approach in (31), this effect is inherently compensated and both bistatic responses are superimposed. The same holds for the above-mentioned pixelwise velocity estimation.

C. Calibration

Typically, the amplitude and phase of an antenna array must be calibrated to enable the estimation of the direction of arrival. Well-known methods use either a target at known or unknown locations in the far field of the antenna array [40]. Compensating array imperfections in the near field is, however, preferred, as calibration in the far field of a large array raises practical issues, such as limited SNR and challenging installation [41]. The following method requires a well reflecting point-like target at a defined location. To improve the calibration quality, i.e., the SNR quality, a sufficiently large number of FMCW chirps need to be sampled and averaged.

This calibration technique has two steps: First, the amplitudes of all monostatic and bistatic responses are equalized. The phases of the monostatic responses are set to zero and the bistatic responses are adjusted to be the complex conjugates of each other, as represented by the phase distortion ϑ_n in (21) and (22). Second, all signals are modulated with the expected phase corresponding to the target position. The ToF values can be calculated with (24) to find the resulting phase shift for a known target location.

As the Fourier transform of the bistatic responses is centered at $f_{\text{center}} = \Delta f + \mu \Delta \tau$ [see (7) and (8)], this value needs to be determined initially; station 2's spectrum is flipped at this value and complex conjugated. Thereafter, the phases of the bistatic signals at the strongest peak before the calibration can be modeled as

$$\begin{aligned}\beta^1_{kmn} &= \arg\left\{\widehat{S}^{2\to 1}_{\text{mix},mn}\left(\arg\max_f\left\{\left|\widehat{S}^{2\to 1}_{\text{mix},mn}(f)\right|\right\}\right)\right\}\\ &= \eta^1_{mn} + \vartheta_{kn} \end{aligned} \tag{32}$$

at the first station and as

$$\begin{aligned}\beta^2_{kmn} &= \arg\left\{\widehat{S}^{1\to 2}_{\text{mix},mn}\left(\arg\max_f\left\{\left|\widehat{S}^{1\to 2}_{\text{mix},mn}(f)\right|\right\}\right)\right\}\\ &= \eta^2_{mn} - \vartheta_{kn} \end{aligned} \tag{33}$$

at the second station, where $\eta^{1/2}_{mn}$ denote the unknown phase values of the corresponding TX-RX combination. Here, the operation in TDM mode causes an unknown phase distortion of ϑ_{kn}, which is a function of the chirp number and the TX element. Therefore, averaging K measurements is not purposeful. To overcome this problem, the mean

$$\beta'_{mn} = \frac{1}{K}\sum_{k=1}^{K} U\left\{\beta^1_{kmn} + \beta^2_{k1n}\right\} = \eta^1_{mn} + \eta^2_{1n} \tag{34}$$

delivers stable results that are independent of the chirp number. Phase unwrapping along the FMCW chirp number avoids ambiguities. These values are used to compensate for the phase difference of the first station, which depends on the RX element number. Accordingly, the average

$$\beta''_{mn} = \frac{1}{K}\sum_{k=1}^{K} U\left\{\beta^1_{k1n} + \beta^2_{kmn}\right\} = \eta^1_{1n} + \eta^2_{mn} \tag{35}$$

provides information on the RX phase difference at the second station. Now, the influence of the RX elements of both stations has been estimated.

After system calibration, the bistatic signals need to be the complex conjugate of each other, which requires the additional calculation of

$$\begin{aligned}\beta'''_{n} &= \frac{1}{KM}\sum_{m=1}^{M}\sum_{k=1}^{K} U\left\{\beta^1_{kmn} - \beta'_{mn} + \beta^2_{kmn} - \beta''_{mn}\right\}\\ &= -\eta^1_{1n} - \eta^2_{1n}. \end{aligned} \tag{36}$$

As the calculations in the last two equations, this result is independent of the chirp number.

The phase calibration values are obtained as

$$\gamma^1_{mn} = \beta'_{mn} + \beta'''_{n} = \eta^1_{mn} - \eta^1_{1n} \tag{37}$$

and

$$\gamma^2_{mn} = \beta''_{mn} = \eta^2_{mn} + \eta^1_{1n}. \tag{38}$$

Therefore, the measured FMCW signals in (30) and (31) need to be corrected by multiplication with the phase terms $\exp(-\gamma^{1/2}_{mn})$. Obviously, all unknown phase values $\eta^{1/2}_{mn}$ are eliminated and both signals are complex conjugated. Finally, the expected phase of each path for a target at a known position is determined by (24) and again multiplied with the input signals.

V. SIMULATION

To assess the performance of the developed algorithms and to get an insight into the expected results of practical measurements in Section VI, simulations are performed. In Section VI-A, the accuracy of the bistatic velocity estimation is evaluated. The enhanced angular accuracy of the coherent combination of two radar stations is evaluated in Section VI-B. To ensure that simulations and measurements are comparable, all radar parameters are selected according to Table I.

A. Widely Separated Radar Stations

To quantify the influence of both phase noise and additive white noise (AWGN) on the bistatic velocity estimation with CFDDS-radar, we simulated a bistatic FMCW radar system composed of two units with a single TX and RX channel (more channels only reduce the influence of AWGN, as PN is virtually the same on all channels).

TABLE I
Simulation and Test Parameters

Symbol	Parameter	Value
f_c	RF carrier frequency	77 GHz
B	RF bandwidth	750 MHz
T_{sw}	Chirp duration	307.2 µs
T_w	Pause between chirps	284.8 µs
T_0	Chirp repetition time	592 µs
μ	Chirp slope	2441 GHz/s
K	Number of chirps	16
N	Number of RX antennas	1, 7 or 8
M	Number of TX antennas	1 or 4
f_s	Sampling frequency	10 MHz
Δf	Frequency offset	1.64 MHz
$\Delta\tau$	Timing offset	75.4 ns
r_{max}	Unambiguous range	112 m
v_{max}	Unambiguous velocity	±0.82 m/s
θ_{max}	Unambiguous azimuth	±90°

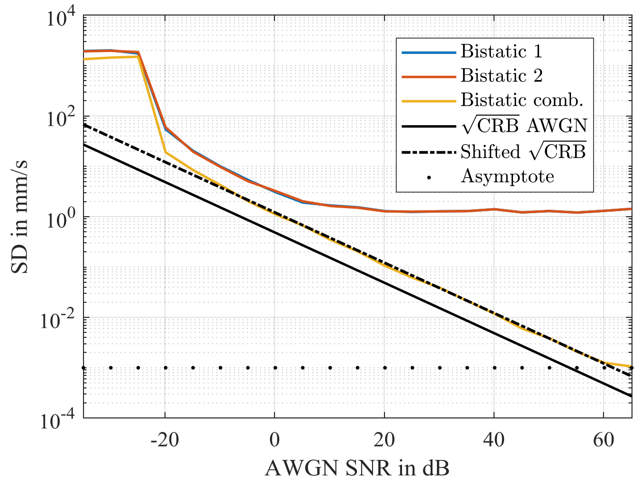


Fig. 6. Simulation results of the CFDDS-radar velocity estimation for a single target at 1.6 m and a velocity of 0.5 m/s. A total of 500 repetitions were computed at 21 equidistant SNR values between -35 and 65 dB. The combination of both bistatic velocity results significantly lowers the SD.

Furthermore, the accuracy of the refinement step in (18) is compared with the individual evaluation in (15) and (16) as a function of the thermal noise SNR. The processing gain after an FFT along fast time dimension with 3072 digital samples is approx. 35 dB. The repetition time of the FMCW chirps was $T_0 = 592$ µs and a chirp sequence is comprised of $K = 16$ chirps. Therefore, the unambiguous range and velocity can be calculated as 112 m and ± 0.82 m/s. If the maximum range is exceeded, monostatic and bistatic responses cannot be separated. A single target at 1.6 m with a constant velocity of 0.5 m/s and angle-independent RCS $\alpha^{1\to 2} = \alpha^{2\to 1}$ was simulated. The phase noise profile was taken from the measurement hardware described in Section VI-A. The measured power spectral density is constant at -74 dBc/Hz for offsets between 100 Hz and 1 MHz and drops off linearly to -84 dBc/Hz for an offset of 3 MHz. The usable dynamic range of the bistatic signals along the fast time dimension was approx. 40 dB and it is strongly limited by uncorrelated PN. Comparable measurements or simulations of the baseband signals and the influence of phase noise in such setups can be found in [8]–[17]. The random phases of each chirp were modeled as uniformly distributed random variables, i.e., $\varphi_k^1, \varphi_k^2 \in U(-\pi, \pi)$. After the corrections in Section III-B2), a 2-D FFT along the range and Doppler dimension was computed. The number of FFT bins was increased by a factor of 8 with zero padding. Thereafter, the exact 2-D peak position was determined by interpolation with a paraboloid.

The standard deviation (SD) of the bistatic velocity estimation is evaluated at 21 equidistant values of the AWGN SNR of a single sample between -35 and 65 dB by Monte-Carlo simulation with 500 repetitions. The results are presented in Fig. 6. The SD after the correction of the unknown phase variables is shown as blue and red curves for stations 1 and 2, respectively. The combination of both bistatic estimates with the refinement step is plotted in yellow. As a lower bound of the estimation accuracy, the square root of the Cramér–Rao bound (CRB) for a pulse sequence radar is depicted as a black line [44]. This bound is not attained [45] as the Hanning window is applied to the signals of both radar units and impairs the velocity estimation accuracy (black dashed line). Moreover, the estimation of the unknown phase values [see (11) and (12)] is corrupted by AWGN. Below an SNR of -20 dB, reliable peak detection from the beat spectra cannot be guaranteed. Between -15 and 0 dB, AWGN is the dominant source of error for the estimation of the unknown phases and the velocity and the precision is roughly doubled by the combination of both bistatic velocities (yellow line) as the available SNR increases. Above an SNR of 5 dB, phase noise predominantly influences the Doppler accuracy of the separate evaluation (red and blue). The SD of the presented method for large SNR values is depicted as a black dotted line at approx. 1 µm/s. The individual results have a lower bound of 1.9 mm/s, which is independent of the AWGN level. Here, the refinement reduces the SD drastically and the resulting performance is comparable to a homodyne radar system, due to the PN suppression. Above 60 dB, the linear approximation in (18) becomes invalid and the accuracy cannot be improved any further with this method. As there are no range-correlation effects on the phase noise of the down-converted signals, the bistatic ranging accuracy is impaired, even for a small target distance. We, therefore, repeated the simulation in [27] for this radar system with known offsets. Comparable to the velocity estimation, the ranging estimation benefits from the estimation refinement if the SNR is above 5 dB. Otherwise, the individual ranging accuracy is bounded by phase noise to 0.9 mm. The computational effort of a single MIMO automotive radar is mainly determined by the complexity of a 3-D FFT. Hence, it is roughly doubled in the case of the bistatic operation in Section III. The presented algorithms

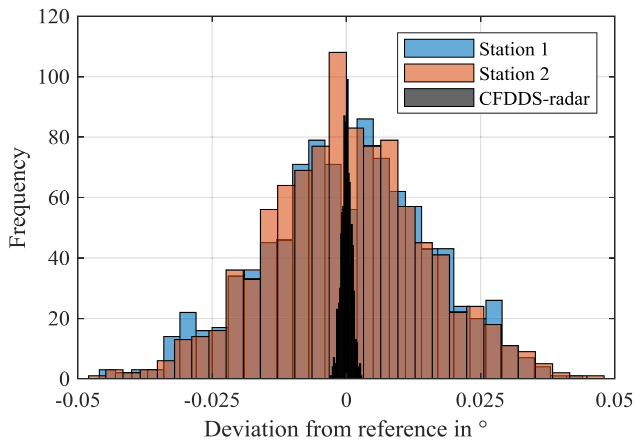


Fig. 7. Histogram of the angular accuracy simulation between $-30°$ and $30°$ with $1°$ steps and 16 repetitions at each position. The SD for stations 1 and 2 is $0.015°$ and can be reduced to $0.001°$ with the coherent combination of both radar stations to a single MIMO array.

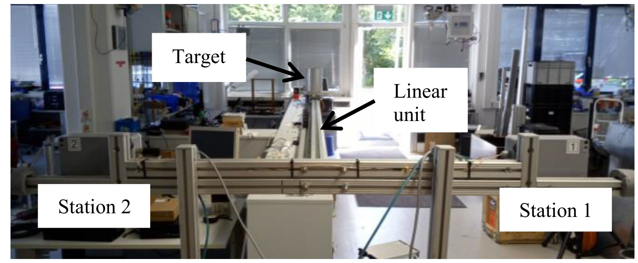


Fig. 8. Bistatic radar setup with a spatial separation of $d_0 = 1.2$ m to assess the accuracy of range and velocity measurements. Both parameters can be set precisely with a linear unit; a metallic cylinder serves as a target.

are based on the estimation of phase values from noisy signals. A mathematical analysis of a related problem in passive MIMO networks is given in [46].

B. Closely Separated Radar Stations

We performed another Monte-Carlo simulation to study the improved angular accuracy of CFDDS-radar with two colocated radar units. All radar parameters are listed in Table I and the calibration is assumed to be perfect. Hence, no artificial phase and amplitude errors or mutual coupling are included in this simulation. Additionally, we assumed the frequency offset $f_{\text{center}} = \Delta f + \mu \Delta \tau$ of the bistatic responses was known. For a fair comparison, we did not use a window function for the single radar units and the full aperture. Fig. 5 shows the array geometry with $N = 4$ TX elements, $M = 7$ RX elements, a gap of $d_2 = 8.6$ cm between the inner RX elements and the corresponding virtual array with 112 elements. The distance between RX and TX elements was $\lambda/2$ and $7\lambda/2$, respectively. A static target distance of 10 m and azimuth angles between $-30°$ and $30°$ were simulated. The angle was sampled at steps of $1°$ and 16 repetitions were computed. After an FFT along the fast time dimension, the AWGN SNR was 27 dB. Again, we modeled the random phases as uniformly distributed random variables. The CRB for the far-field detection for an arbitrary linear array is given in [47]. While this bound is valid for a single radar unit array, it is invalid for the complete aperture, because an additional phase correction is required, and detection by the spatially matched filter in the array near field. For a single radar unit with a MIMO aperture consisting of 28 antenna elements with $\lambda/2$ interelement spacing, the (angular dependent) lower bound of the angular estimation accuracy is approx. $0.014°$.

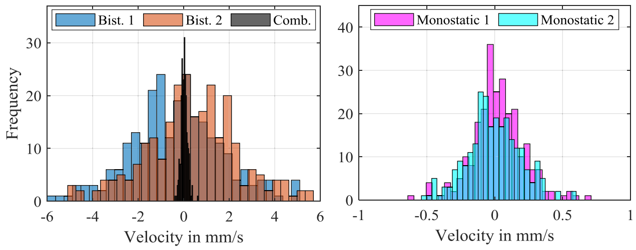
Fig. 7 shows the histogram of the deviation of the estimation results from the reference angle. The reconstruction was computed with the spatially matched filter in (23) and (31) for the single radar stations and the full array, respectively. In both cases, the bin size was 5 mm in x and y dimension and the precise target location was then established by the 2-D interpolation. Blue and red bars indicate the results for stations 1 and 2, which are normally distributed with an SD of $0.015°$. With the aid of CFDDS-radar, the SD is reduced to $0.001°$ and individual results are depicted as black bars, which show a significantly lower deviation from the reference angle. The computational complexity is mainly determined by the spatially matched filter reconstruction in (31) and the phase correction in (26). For each pixel, $6 \cdot M \cdot N$ complex multiplications are required.

VI. MEASUREMENTS

In this section, we present coherent measurement results with two separated FMCW radar units and CFDDS-radar. We assess how well ranging and velocity are estimated by our proposed technique in a laboratory setting where the distance d_0 between stations 1 and 2 is 1.2 m. We evaluate the enhanced peak level for a complex target, as well, in the same test. The angular accuracy is quantified and an improved resolution is demonstrated with this fully coherent approach where the spatial separation between the stations is small.

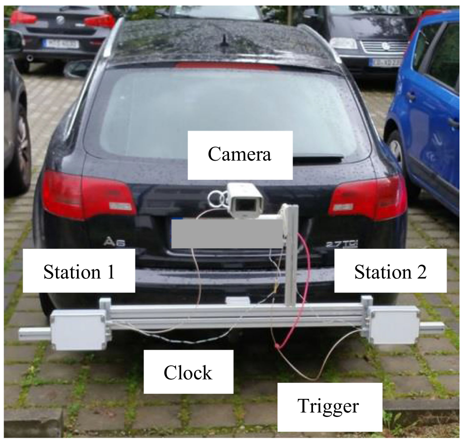
A. Description of the Measurement Hardware

For all measurements, we used the 77 GHz evaluation platform "Radarbook," manufactured by INRAS, as stations 1 and 2 and the distance in between was either $d_0 = 1.2$ m or $d_2 = 8.6$ cm. These FMCW MIMO radar systems with eight RX and four TX elements were connected by a coaxial cable, which transmits a trigger signal from station 1 (master unit) to station 2 (slave unit) to start a chirp sequence. The 80 MHz system clock of the first station was distributed by a twisted pair cable fed by a line driver and the frequency of a local voltage-controlled crystal oscillator at the second station was adjusted to this reference. Both cables of the bistatic radar setup in Fig. 10(a) are marked. A timing offset of 75.4 ns is comprised of the transmission time and an additional delay caused by digital processing in the slave unit. The second radar was operated with a constant frequency offset of 1.64 MHz to enable the monostatic and bistatic responses to be separated in the beat spectrum after range compression. Both offsets were determined by sequential

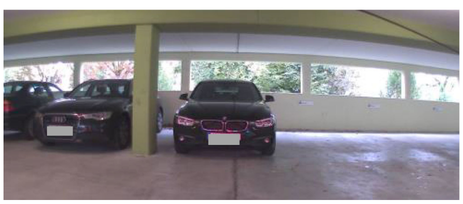


(a) (b)

Fig. 9. Velocity measurement results in a static scenario with 50 repetitions at five positions between 1.5 and 5.5 m. The combination of both bistatic responses with CFDDS-radar reduces the SD from 2 to 0.14 mm/s. (a) and (b) This is comparable to the monostatic SD of 0.18 mm/s and 0.19 mm/s for stations 1 and 2, respectively.



(a)



(b)

Fig. 10. Radar setup with a spatial separation of $d_0 = 1.2$ m. (a) 80 MHz system clock was shared with a twisted pair cable and the trigger signal was transmitted by a coaxial cable. A camera serves as an optical reference. (b) Measurement scenario on a parking deck with static cars.

up- and downchirps and the method in [28] prior to the tests.

As multiplexing cannot be carried out with FDM, one TX element was used for all dynamic measurements in Section VI-B. We used all four TX elements to create the full aperture in Fig. 5, but only seven RX channels to create an aperture with no redundant elements. In this experiment, we investigated a static scenario and operated both radar units with TDM. We selected relevant parameters as per Table I. The ADC data from both units were transmitted to a PC via IEEE 802.3 wired Ethernet and processed offline.

B. Widely Separated Radar Stations

In this section, the accuracy of ranging and velocity is assessed, and a practical test will be explained.

1) *Evaluation of the Velocity and Ranging Accuracy:* The bistatic velocity cannot be evaluated without additional processing steps due to the random phase distortions. To assess the performance of the CFDDS-radar algorithm in Section III, we conducted a laboratory experiment. A photograph of the measurement setup with two radar stations with a spatial separation of $d_0 = 1.2$ m is shown in Fig. 8. A metallic cylinder on a linear unit serves as a point-like target at precise locations or with a defined velocity. As such, the cylinder is a strong isolated target and the multipath caused by the linear rail and other objects is weak. The SNR for this object was roughly 20 dB for a target distance of 4.5 m.

To avoid the effects of multiple reflections and scattering at objects, first test was done with a static target at five different distances of $y \in \{1.5; 2.5; \ldots; 5.5\}$ m from the center of the radar stations. At each position, the measurement was repeated 50 times, which makes a total of 250 measurements. Results of the velocity estimation are shown in Fig. 9, where the bistatic measurement is depicted in (a). The evaluation after a 2-D FFT of (15) and (16) after the correction of the random phase values is shown in blue and red. The average SD for all positions in this evaluation is 2 mm/s. Again, we used a Hanning window in both dimensions. We also used zero padding and a 2-D interpolation (see Section V). After refining (i.e., averaging) the latter results, the average SD decreases to 0.14 mm/s. This result is plotted in black. It is notable for its significantly reduced spread. The monostatic velocity estimation is depicted in Fig. 9(b) for comparison purposes. The SDs for stations 1 and 2 are 0.18 mm/s and 0.19 mm/s, respectively. These results are less accurate than with the combination of both bistatic results, as the available SNR after the refinement is larger than with the individual estimation. This result is in good accordance with the simulation results in Fig. 6 for an SNR of 20 dB. The measured and simulated noise levels of the spectra are also comparable.

Additionally, the ranging SD was evaluated with data from this measurement. Here, the resulting SDs are 1.3 mm, 1.2 mm, and 0.11 mm for the bistatic responses of station 1, station 2, and the bistatic combination, respectively. Suppressing the phase noise along the fast time dimension reduces the SD by a factor of approx. 11. The latter result is again lower than the SDs of the monostatic responses, which we determined to be 0.17 mm and 0.15 mm for stations 1 and 2, respectively, given the higher SNR.

We conducted another measurement with this setup to evaluate the root-mean-square error (RMSE) and the bias of this method. To this end, we chose the colocated setup in Fig. 5 with a small spatial separation to ensure that the measured radial velocity of both radar stations is almost the same. We preset the linear unit velocity to $v_{\mathrm{ref}} \in \{0.1; 0.5\}$ m/s and recorded the ADC data from 46 chirp sequences. The results of the bistatic and monostatic target responses were highly correlated and the measured RMSE was between 2 and 7 mm/s, due to multipath propagation. The bias was between -2 and 5 mm/s, regardless of the specific method.

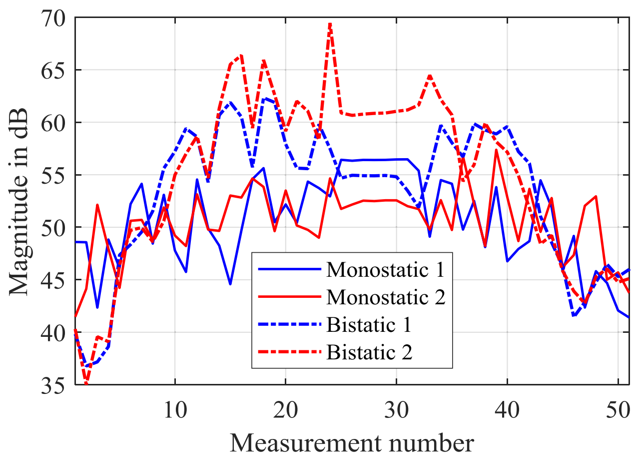


Fig. 11. Measurement results for a target with angle-dependent RCS at 51 positions (see Fig. 10). The ego vehicle approached a car to 1.3 m starting from 5.4 m. The AWGN noise floor after an FFT along fast time dimension was at −5 dB.

In summary, the accuracy of the bistatic measurements with CFDDS-radar after both bistatic responses are combined is comparable to the accuracy of a monostatic radar system.

2) *RCS Measurement in a Practical Scenario:* In this test, a car was equipped with the bistatic radar setup, as depicted in Fig. 10(a). A camera was used as an optical reference and all devices were attached on a robust aluminum rail. Given the vehicle is moving and multiplexing can be realized via TDM only, the RF signals are radiated by a single TX element and the phase offsets of all eight RX channels were corrected by calibration. This vehicle started at 5.4 m distance to a parked car [as shown in Fig. 10(b)] until it was 1.3 m away from it and returned to its starting position. We took measurements at 51 points along the car's linear trajectory. Besides the two monostatic responses, we used the bistatic mixing products to compute four range-Doppler-azimuth diagrams, where the latter signals were processed with the algorithm in Section III-B2). We omitted the refinement step to obtain information on the RCS along four different signal paths. In this experiment, the AWGN noise floor was approx. at −5 dB after an FFT along the fast time dimension.

The maxima of the 3-D FFT with CFDDS-radar are plotted in Fig. 11 for all target responses. These values are obtained after zero padding with a factor of 16 in all three dimensions to keep the discretization error sufficiently low. A Hanning window was used in this test. This car is not a point-like target, given its curvature, and the RCS depends on the perspective. The detected peak levels of the bistatic responses are higher than the peak levels for monostatic signals for short distances, especially. The measured mean peak levels, which are functions of $\alpha^{1\to1}$, $\alpha^{2\to2}$, $\alpha^{2\to1}$, and $\alpha^{1\to2}$, are 50.4 dB, 50.4 dB, 53.1 dB, and 54.6 dB, respectively. Hence, the SNR and the visibility of complex objects can be increased using all the various responses. Fig. 11 also shows that both bistatic responses are correlated but the measured SNRs are different, as they depend on the exact antenna positions.

C. Closely Separated Radar Stations

This section demonstrates the fully coherent operation mode, i.e., two separate apertures combined into a single array.

1) *Measurement of the Angular Estimation Accuracy:* We compared the increased angular estimation accuracy of the two coherently combined radar stations with the measurement results of the individual radar units at 61 uniformly distributed test angles between −30° and 30° in a practical trial. To this end, both devices were fixed on a metal plate, as shown in Fig. 12(a). We used the aperture from Fig. 5 here, as well. Defined angular values were set precisely with a rotary table [see Fig. 12(b)]. Much more precise results can be achieved with this configuration than was possible with a dynamic target. The radar units were covered with a plastic radome above the rotary table. The metallic cylinder was placed 10 m away from the radar stations, as shown in Fig. 12(c). The system was calibrated at 0° prior to the measurement with the method in Section IV-C to eliminate all phase offsets. All channel amplitudes were equalized, as well.

Fig. 13 shows a histogram of the deviation from the reference, where the angular estimations with stations 1 and 2 are shown in blue and red, respectively. This is the evaluation of the monostatic responses, i.e., the signals that can be measured with one device. The RF signals were radiated by 4 TX elements and received by 7 RX channels and the resulting virtual array is comprised of 28 elements. The test was repeated 16× for each angle in 1° steps, which makes a total of 976 measurements to calculate statistical results. The RMSEs of all 976 samples were computed as 0.136° and 0.108° for stations 1 and 2, respectively. Black bars in Fig. 13 indicate the results of the coherent measurement with all 112 channels and CFDDS-radar. The RMSE is reduced to 0.012° by a factor of 11 and 9 for stations 1 and 2, respectively, by coherently processing all monostatic and bistatic responses. The SD of the angular estimation at each position was, on average, 0.025° for station 1 and 0.020° for station 2. It can be enhanced by a factor of roughly 13 and 17 to 0.0015° by CFDDS-radar with a full aperture. This improvement is in accordance with the simulation in Section V-B. Different error sources are typically encountered. They include mutual coupling, inaccuracies of the element positions of both radar stations, the influence of the radome and the gap in between, an additional reflection on the ground or reference inaccuracies, which might explain the RMSE discrepancies, and the resulting SDs.

2) *Measurement of the Enhanced Resolution Capabilities:* The last test demonstrates the higher resolution with two adjacent targets by means of coherent processing. Fig. 14 pictures the measurement setup with a single isolated target at the position $\mathbf{x}_{\mathrm{T},1} = (0, 2.5)^{\mathrm{T}}$ m and two metal poles at a radial distance of approx. 1.75 m and an azimuth angle of 14° with a spatial separation of 7 cm in between. Given that both poles have the same RCS, the resulting amplitudes after the reconstruction are also expected to be similar. The

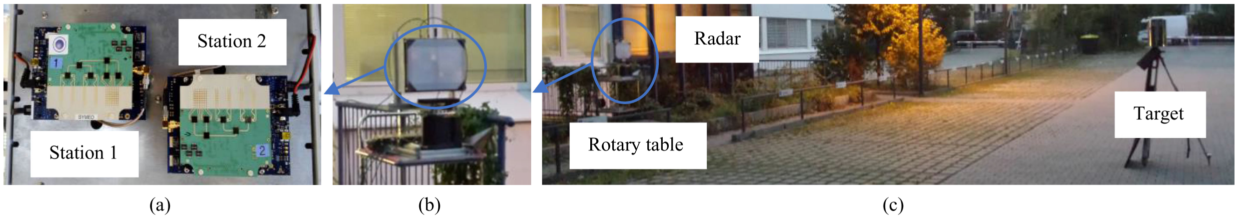


Fig. 12. Measurement setup with a radar mounted on a rotary table for precise angular reference. A metal cylinder at 10 m serves as a point-like target. (a) Stations 1 and 2 are close to each other. (b) Rotary table is underneath the radar stations, which are situated behind a plastic cover. (c) An overview of the setup.

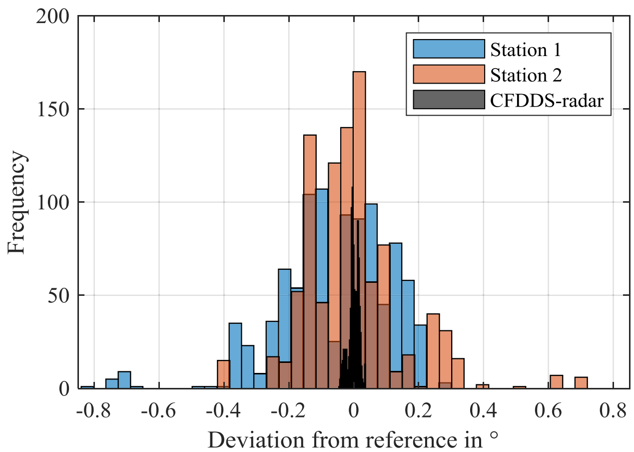


Fig. 13. Histogram of the angular accuracy for measuring a target at 10 m. The individual RMSEs for stations 1 and 2 of 0.136° and 0.108°, respectively, are reduced to 0.012° by a fully coherent operation. The SD can be reduced from 0.025° and 0.020° at stations 1 and 2, respectively, to 0.0015°.

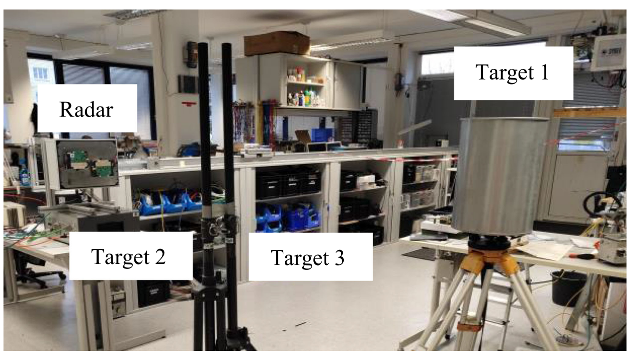


Fig. 14. Radar setup for the resolution measurement with an isolated target at $\mathbf{x}_{T,1} = (0, 2.5)^T$ m and two adjacent targets with a spatial separation of 7 cm at a radial distance of approx. 1.75 m.

required corrections in Section IV-B are performed with the metallic cylinder response.

All spatially matched filter reconstruction results of the measurement data without a window function are depicted in Fig. 15 where targets are marked by black crosses. A comparison of the point spread function (PSF) of a single radar in (a) and the full interferogram with CFDDS-radar in (b) shows that the angular resolution was improved by a factor of 15 from 9.7° to 0.64°. To keep the magnitude of all sidelobes at least 6 dB below the main lobe, the gap between both radar units was again chosen as 8.6 cm. The two metal poles cannot be resolved by a single radar due to the limited aperture (c). Both targets exhibit constructive interference and the reconstruction forms a single peak at this location. An interferogram of all available target responses reveals two distinct maxima with a spatial separation of 7.05 cm, which is very close to the expected distance of 7 cm (d). As complex superposition of both target responses, this interferogram depends on the difference between the signal paths. In this scene, the amplitude of the target on the left was 1 dB below the target on the right.

VII. DISCUSSION

Additional steps are necessary to enable the use of CFDDS-radar in automotive radar networks. Above all, reliable detection in arbitrary scenarios without the requirement of a strong isolated target is feasible with a common TX and RX element (TRX element) at each station. By that, the measured channel response is identical in both directions and specific targets can be ignored. If a TRX element is not available, processing-based coherent radar networks with the presented method might still be realized. To this end, the influence of the geometry on the initial phase must be compensated, for instance, by the generic correction presented in Section IV.B.

All TX elements will need to be used simultaneously to properly detect moving targets with a MIMO array. This can be accomplished with FDM instead of TDM. The unambiguous Doppler of a few meter per second must be increased by reducing the chirp repetition time to cope with the expected velocities in typical traffic situations. Furthermore, the SLL of the PSF with the full array in this article was at −6 dB. This should be enhanced by modifying the array configuration or by using compressed sensing techniques for reconstruction [48]. Practical applications also require a cover, i.e., the installation of the radar units behind the bumper or a dielectric lens. Additional reflections impair the angle estimation due to errors in amplitudes and phases and increased the mutual coupling. To avoid demanding manual calibration, the self-calibration method in [49] can be used. To avoid all physical wiring, the method in [13] might be extended to MIMO radar systems.

Previous studies on the processing-based coherent distributed radar with a comparable setup have been conducted at a carrier frequency of 122 GHz with 1 GHz RF bandwidth in [12]. The evaluation of a sequence of 30 FMCW chirps

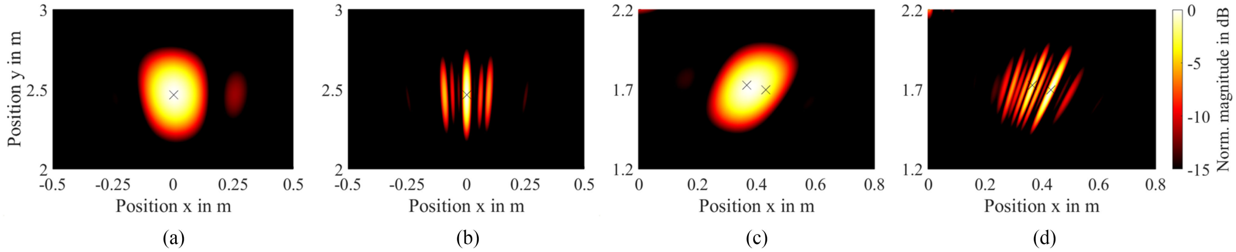


Fig. 15. Results of the resolution measurement with (a) and (b) single target and (c) and (d) two targets with a spatial separation of 7 cm. The PSFs are shown for the monostatic setup in (a) and (c) and the bistatic setup with CFDDS-radar in (b) and (d). The two targets cannot be resolved due to the limited aperture of a single radar (c). The two largest peaks, which appear at the target positions in (d), can be detected with the CFDDS-radar reconstruction.

results in a precision of the measured speed lower than 1 mm/s, which is in good accordance with the measured SD of 0.14 mm/s in this work. Furthermore, the measured ranging SD of 0.11 mm by combining all bistatic responses is comparable to the 0.63 mm in [12]. Due to the large virtual aperture in this work, the measured SD of 0.0015° for the angular estimation is much lower than 0.18° that is measured with a virtual array with four elements in [9].

VIII. CONCLUSION

On the one hand, we have described CFDDS-radar for widely separated FMCW radar units without a common RF carrier frequency in this article. Despite the unknown initial phases and the influence of phase noise, the resulting accuracy of bistatic ranging and velocity measurements is comparable to the performance of a homodyne radar system. We have also shown that the SNR for a complex target can be enhanced by employing different perspectives.

On the other hand, a large virtual array can be synthesized with the colocated MIMO setup and the target response can be reconstructed with a spatially matched filter in the near field of the aperture. Hence, the resulting accuracy of the angular measurements has been improved. Additionally, using experimental data, we have demonstrated the higher resolution achievable by fully processing all available signals coherently.

The presented CFDDS-radar concept and results pave the way for sophisticated but cost-efficient coherent radar networks and for tremendously improved performance of future automotive radar systems.

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